OpenShift Virtualization – Deployment & VM Lifecycle

1. Overview

OpenShift Virtualization (CNV) enables running Virtual Machines (VMs) on OpenShift using KubeVirt. This document describes full deployment of CNV on OpenShift and running a VM successfully.

2. Architecture Overview

Stack

- OpenShift Container Platform (OCP)
- OpenShift Virtualization (KubeVirt)
- OpenShift Data Foundation (ODF)

Flow

User → OpenShift Console → VM Definition → virt-controller → virt-launcher pod → KVM (node)

3. Prerequisites

Cluster Requirements

- OpenShift 4.x installed and running
- Worker/master nodes in Ready state
- Cluster-admin access

Storage

- ODF installed
- StorageClass available (Ceph RBD recommended)

Critical Requirement

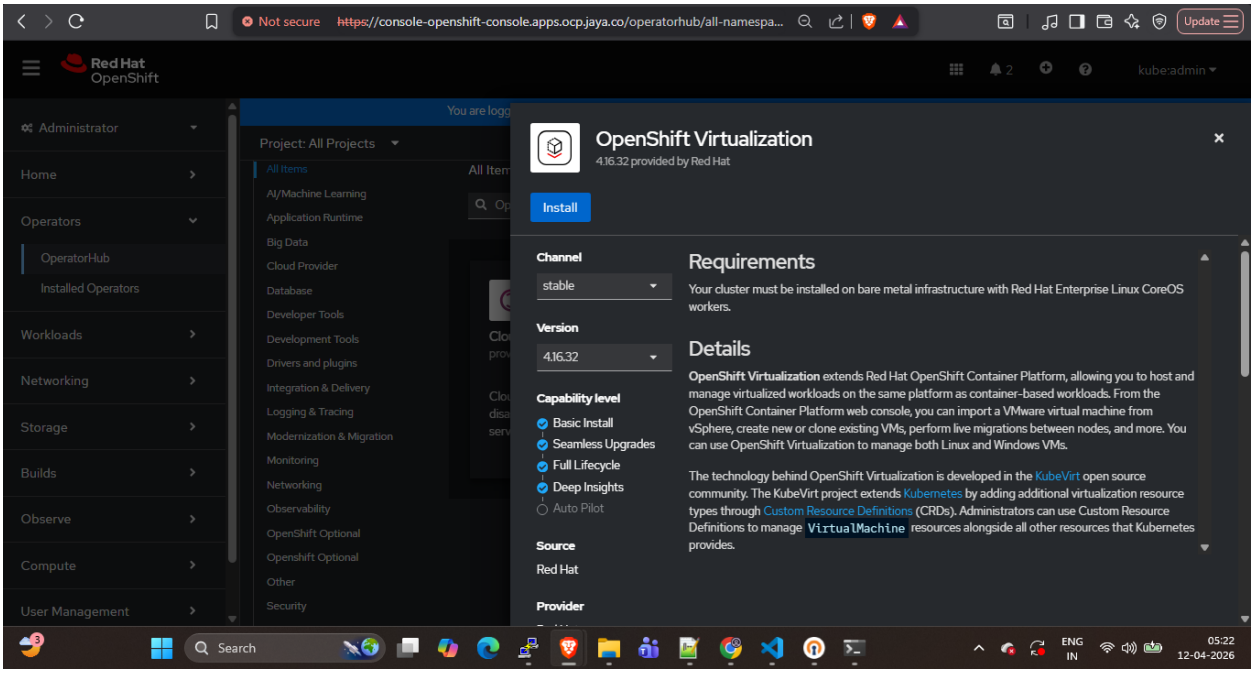
- Hardware virtualization (VT-x / AMD-V) must be enabled on nodes

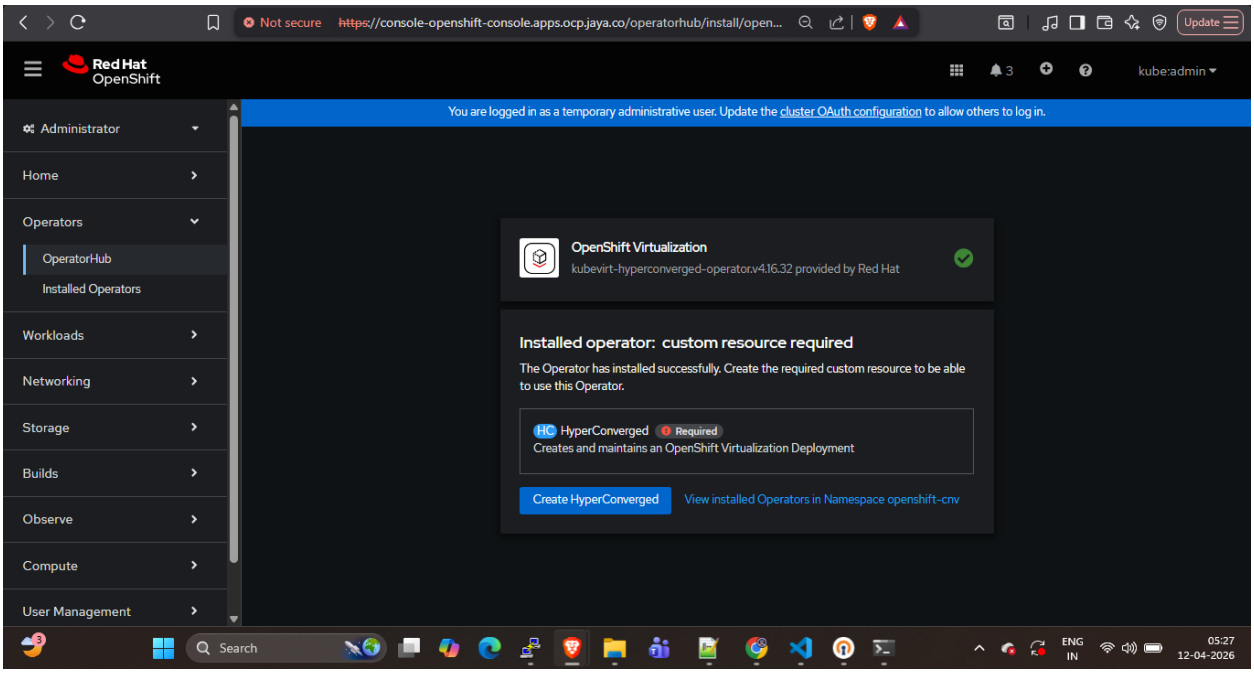
4. Installing OpenShift Virtualization (CNV)

4.1 Install Operator

Go to:

OperatorHub → OpenShift Virtualization





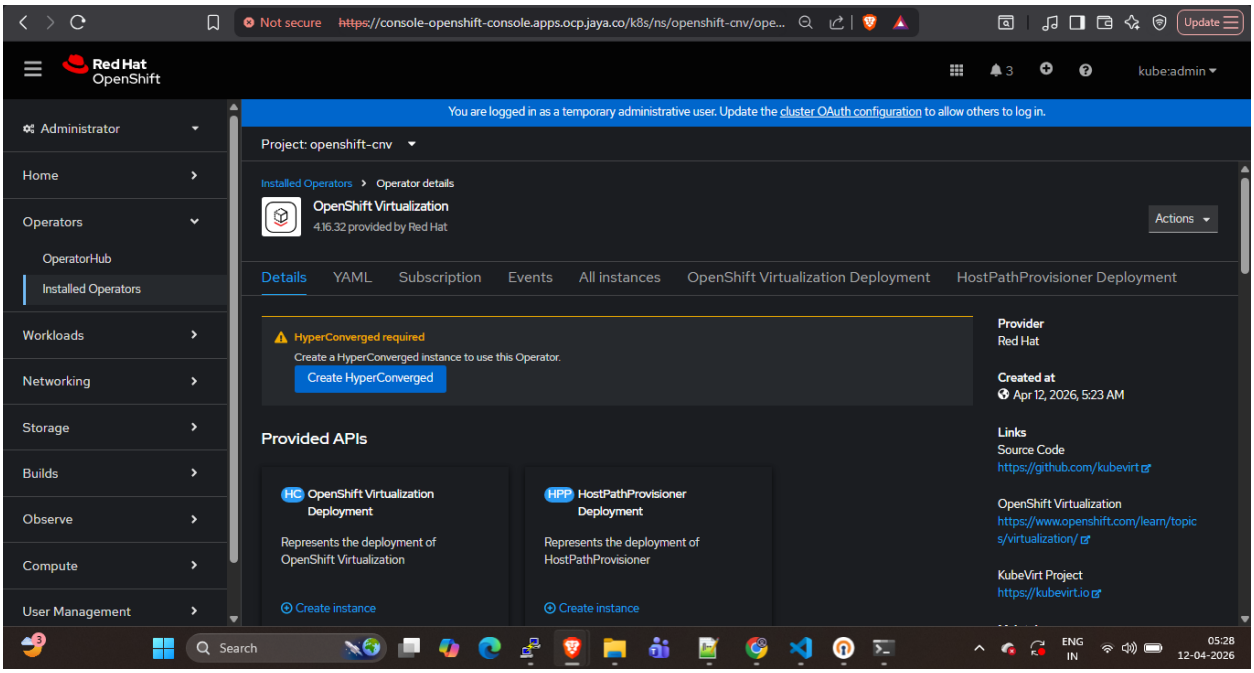
Install:

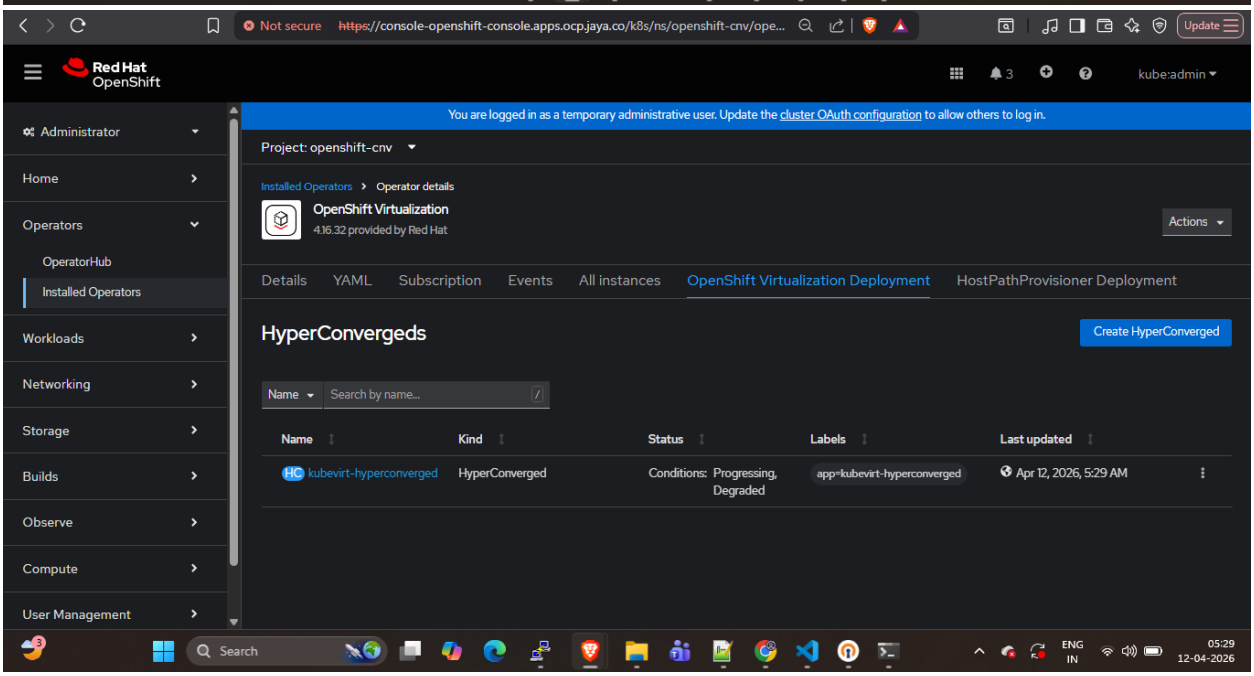
- OpenShift Virtualization Operator

4.2 Create HyperConverged (HC) Instance

Go to:

Installed Operators → OpenShift Virtualization → Create HyperConverged





Apply default configuration:

spec: {}

This deploys:

- virt-operator
- virt-controller
- virt-handler

- CDI
- SSP
- Networking components

4.3 Verify Installation

oc get pods -n openshift-cnv

```
[root@bastion install]# oc get pods -n openshift-cnv -w
NAME                                                READY   STATUS    RESTARTS        AGE
aaq-operator-7455b9bcff-bdn4p                       1/1     Running   3 (13m ago)     28m
bridge-marker-dd48g                                 1/1     Running   0               24m
bridge-marker-gjqqf                                 1/1     Running   0               24m
bridge-marker-jf4c7                                 1/1     Running   0               24m
cdi-apiserver-584d59f965-pjzn2                      1/1     Running   0               24m
cdi-deployment-5b5d7886c-ct6n6                      1/1     Running   3 (12m ago)     24m
cdi-operator-66b885d546-msfwh                       1/1     Running   3 (13m ago)     28m
cdi-uploadproxy-6d88777ddf-jdsvr                    1/1     Running   0               24m
cluster-network-addons-operator-787d64f8bb-g74mc    2/2     Running   0               29m
hco-operator-67d7f4d995-xs6zl                       1/1     Running   4 (14m ago)     29m
hco-webhook-fb646b74d-n7g89                         1/1     Running   0               29m
hostpath-provisioner-operator-9659fccdc-q6n8m       1/1     Running   4 (10m ago)     28m
hyperconverged-cluster-cli-download-65c778b7f9-znc7p 1/1    Running   0               29m
kube-cni-linux-bridge-plugin-2r2jz                  1/1     Running   0               24m
kube-cni-linux-bridge-plugin-dq44x                  1/1     Running   0               24m
kube-cni-linux-bridge-plugin-ngtwk                  1/1     Running   0               24m
kubemacpool-cert-manager-6cdd458b4b-xjmgv           1/1     Running   0               24m
kubemacpool-mac-controller-manager-675f96bfcd-tq787 2/2     Running   0               24m
kubevirt-apiserver-proxy-54b9854945-2g5bg           1/1     Running   0               23m
kubevirt-apiserver-proxy-54b9854945-d9gr8           1/1     Running   0               23m
kubevirt-console-plugin-7ff8c458bd-8rqss            1/1     Running   0               23m
kubevirt-console-plugin-7ff8c458bd-wcxrn            1/1     Running   0               23m
mtq-operator-686dbc55c-dmhmz                        1/1     Running   2 (13m ago)     28m
ssp-operator-5fb647b7b8-wqjt4                       1/1     Running   3 (13m ago)     29m
virt-api-748bdbc496-hnlpt                           1/1     Running   0               23m
virt-api-748bdbc496-lf5kb                           1/1     Running   0               23m
virt-controller-67d4545458-b498w                    1/1     Running   3 (8m47s ago)   21m
virt-controller-67d4545458-vwq52                    1/1     Running   5 (13m ago)     21m
virt-exportproxy-7b66f899b8-cq9r2                   1/1     Running   0               21m
virt-exportproxy-7b66f899b8-qv68b                   1/1     Running   0               21m
virt-handler-717b8                                  1/1     Running   0               21m
virt-handler-14k72                                  1/1     Running   0               21m
virt-handler-lr5xc                                  1/1     Running   0               21m
virt-operator-c99fd84cd-gg6wj                       1/1     Running   4 (12m ago)     28m
virt-operator-c99fd84cd-mj6vd                       1/1     Running   2 (19m ago)     28m
virt-template-validator-6677bf57c9-ddr5m            1/1     Running   0               23m
virt-template-validator-6677bf57c9-wb2qg            1/1     Running   0               23m
```

5. Storage Configuration

Default StorageClass

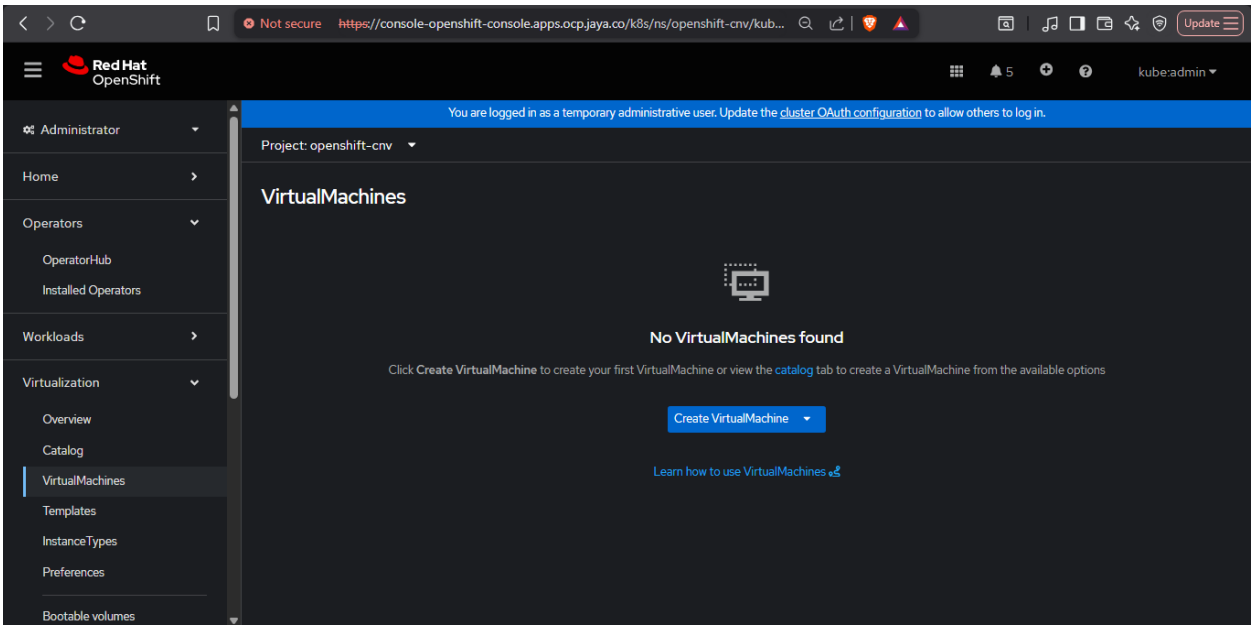
Use ODF Ceph RBD: ocs-storagecluster-ceph-rbd

This is used for VM disk provisioning.

6. VM Creation

6.1 Navigate

Virtualization → VirtualMachines → Create VirtualMachine

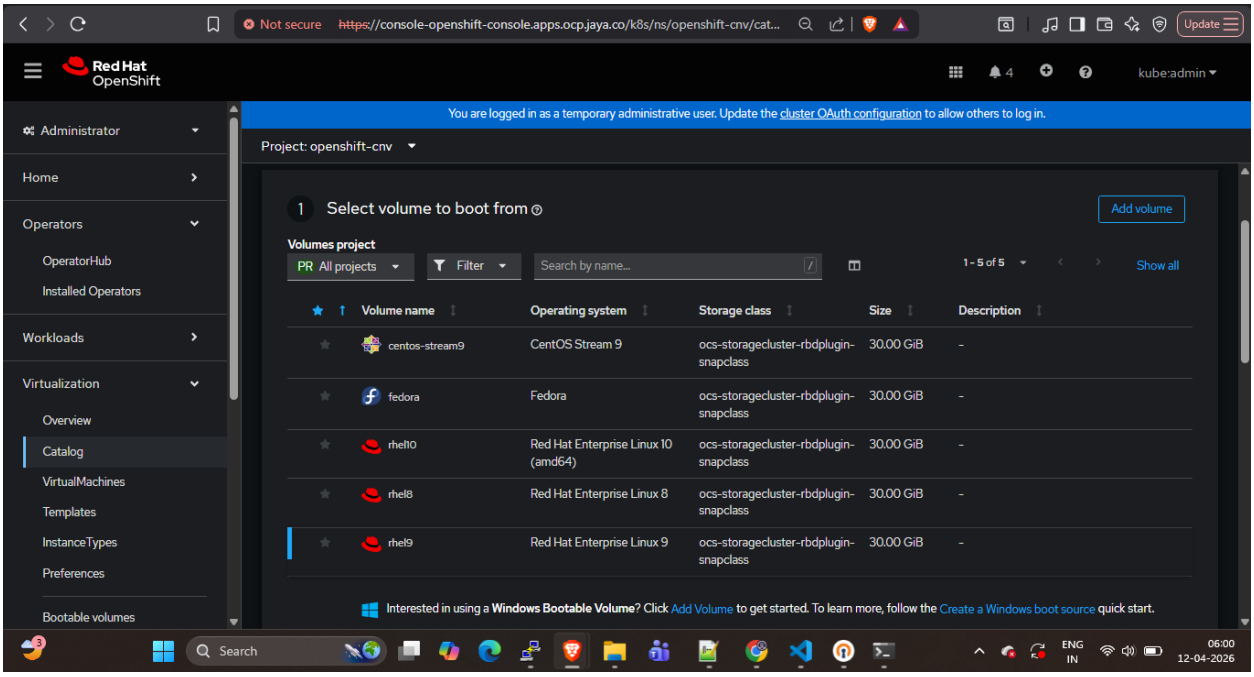


6.2 VM Configuration

Recommended settings:

Parameter	Value
Name	rhel-9-vm
OS	RHEL 9
CPU	1–2 cores
Memory	2–4 GB

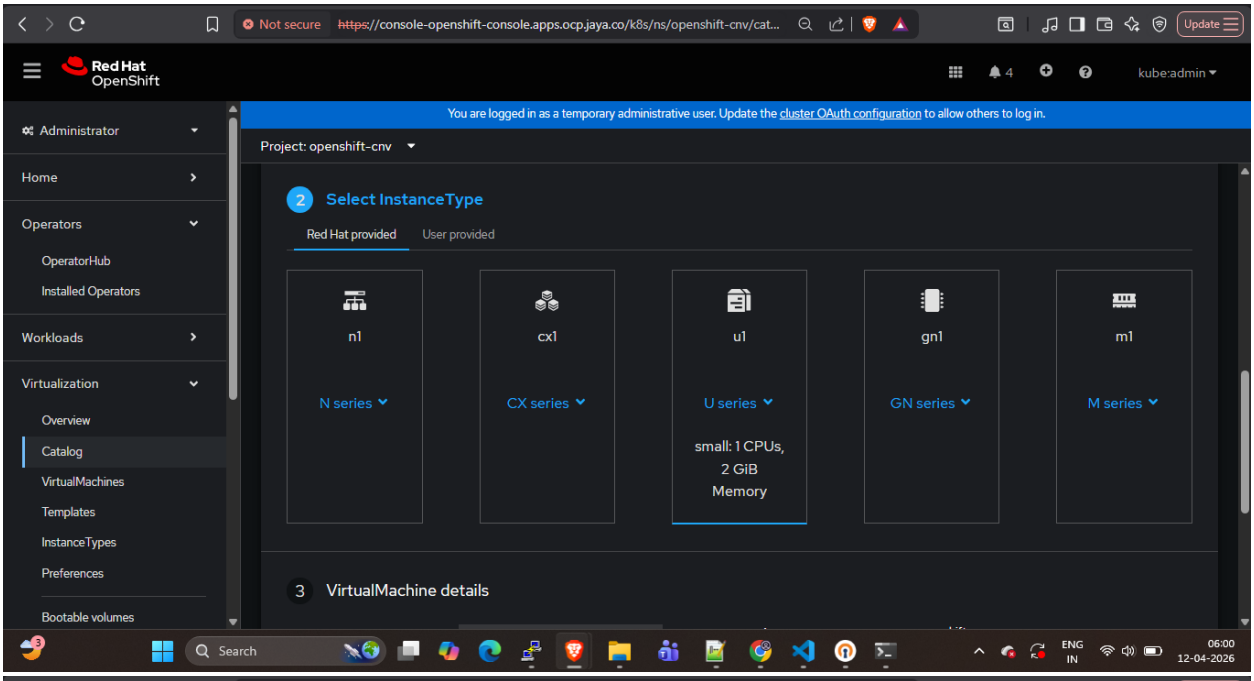
Disk	20–30 GB
StorageClass	ocs-storagecluster-ceph-rbd

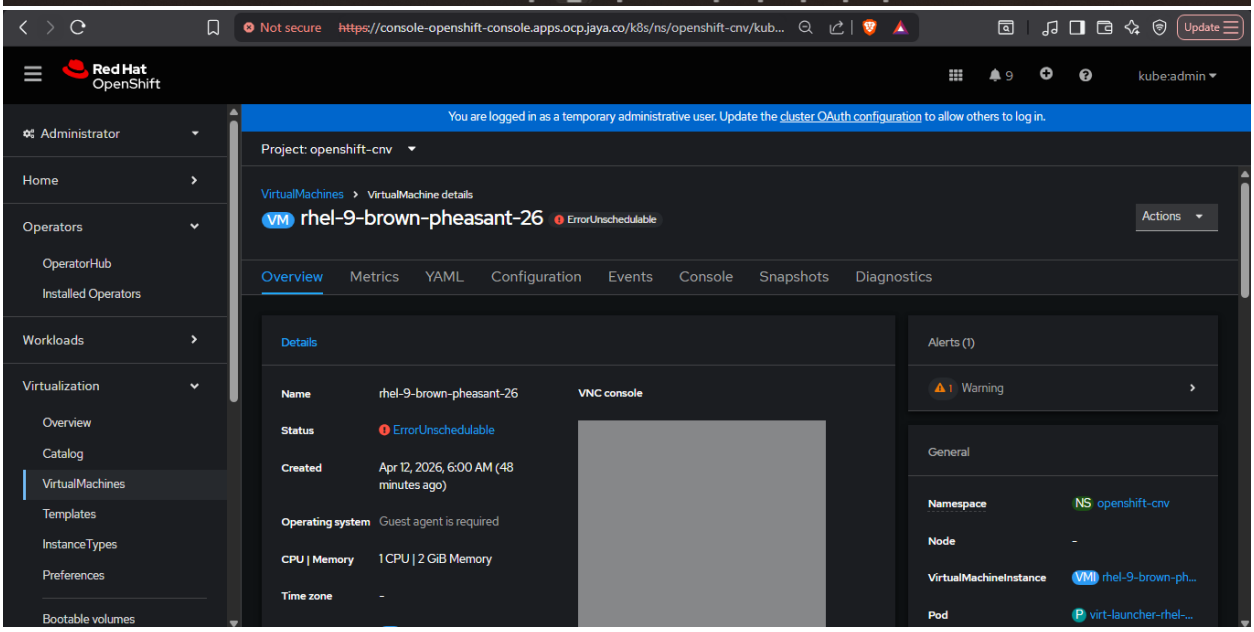


6.5 Create VM

Click:

Create → Start VM





7. VM Lifecycle

Step 1: VM Object Created

- Kubernetes VM CR is created

Step 2: VMI Created

- VirtualMachineInstance is generated

Step 3: Scheduling

- virt-controller assigns node

Step 4: Execution

- virt-launcher pod runs VM using KVM

8. Validation Commands

Check VM

```
oc get vm -n openshift-cnv
NAME                        AGE   STATUS               READY
rhel-9-brown-pheasant-26    48m   ErrorUnschedulable   False
```

Check VMI

```
oc get vmi -n openshift-cnv
NAME                        AGE   PHASE        IP    NODENAME   READY
rhel-9-brown-pheasant-26    48m   Scheduling                    False
```

Check Pod

```
oc get pods -n openshift-cnv | grep virt-launcher
virt-launcher-rhel-9-brown-pheasant-26-xw6td          0/1     Pending   0
48m
```

Get Node VM capability

```
egrep -c '(vmx|svm)' /proc/cpuinfo
0
```

9. Observed Issue in Lab Environment

Symptom

- VM stuck in Scheduling state
- VMI not reaching Running phase
- virt-launcher pod not scheduled

Error

ErrorUnschedulable

10. Root Cause Analysis

Finding

- No hardware virtualization available on nodes

Verification

```
egrep -c '(vmx|svm)' /proc/cpuinfo
```

Output:

0

Impact

- KVM not available
- No VM-capable node found
- Scheduler fails to place VM

11. Final Root Cause

OpenShift Virtualization requires hardware-assisted virtualization (KVM).

In this environment:

- VMware ESXi does NOT expose VT-x/AMD-V to guest nodes
- Nested virtualization is disabled

Result:

- CNV is installed successfully
- VM execution fails completely

12. Fix Requirement (ESXi Level)

To enable VM functionality:

ESXi Configuration

On OpenShift node VM:

- Power OFF VM
- Enable:
  - "Expose hardware assisted virtualization to the guest OS"
  - OR "Virtualize Intel VT-x/EPT or AMD-V/RVI"

After enabling:

- Reboot node
- Verify:

```
egrep -c '(vmx|svm)' /proc/cpuinfo
```

Expected:

- Value > 0

13. Post-Fix Validation

```
oc get vmi -n openshift-cnv
```

Expected:

- VMI = Running
- IP assigned

```
oc get vm -n openshift-cnv
```

Expected:

- VM = Running

14. Conclusion

OpenShift Virtualization deployment is successful when:

- Operators are installed
- HyperConverged CR is created
- Storage is configured
- Nodes support hardware virtualization

In this setup, VM failure is due to ESXi not enabling nested virtualization.

Final Note

In VMware ESXi-based environments, nested virtualization must be explicitly enabled. Without it, OpenShift Virtualization can be installed successfully but cannot run any virtual machines.

```
[root@bastion install]# egrep -c '(vmx|svm)' /proc/cpuinfo
0
[root@bastion install]#
```